

2(a)(i)	AAbb, Aabb, aaBB, aaBb ;					1
2(a)(ii)		<i>Alfresco</i>	<i>Di Nizza</i>	<i>or</i>	<i>Di Nizza</i>	4
	parent genotypes	aabb	Aabb		aaBb ;	
	gametes	ab	Ab ab		aB ab ;	
	offspring genotypes	aabb	Aabb		aaBb ;	
	offspring phenotypes	long	spherical		spherical ;	
2(a)(iii)	random / independent, assortment ; metaphase 1 ;					2
2(b)(i)	<i>any two from:</i> 1 gourds might become invasive ; 2 more competition from gourds might reduce food production ; 3 might harm gourd genetic diversity ; 4 resistant gourds might have, unknown / adverse, effects on ecosystem ;					2

Question	Answer	Marks
2(b)(ii)	any two from: 1 resistance gene makes hybrids more susceptible to <i>Erwinia</i> ; 2 gene flow less of a problem than predicted (for this GMO) ; 3 GM squashes are 'safe' to grow ; 4 AVP ; e.g. could still be a problem in areas lacking, beetle / <i>Erwinia</i>	2
2(b)(iii)	insecticides used to, kill beetle <u>vector</u> / prevent beetles spreading <i>Erwinia</i> or beetles may only feed on hybrids ;	1